Chain Rule

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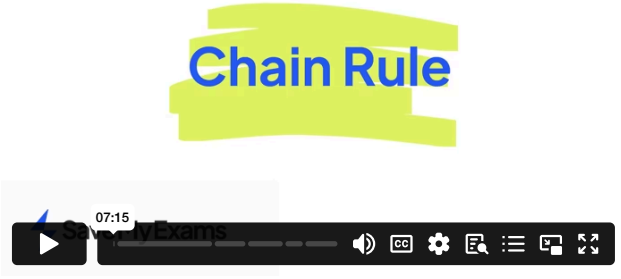

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Help me understand

Chain rule

What is the chain rule?

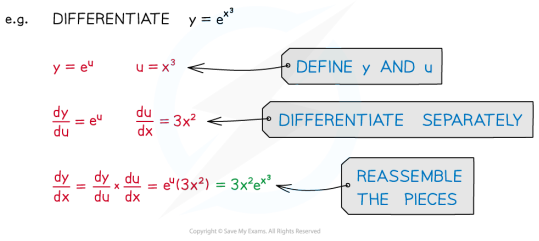
- The chain rule is a formula that allows you to differentiate **composite functions**
- If ***y*** is a function of ***u***, and ***u*** is a function of ***x***, then the **chain rule** tells us that:

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

- Note that because ***u*** is a function of ***x***, ***y*** is also a function of ***x***
- In **function notation**, if $y = f(g(x))$ then the chain rule can be written as:

$$\frac{dy}{dx} = f'(g(x))g'(x)$$

This allows us to differentiate more complicated expressions:



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Got it

$f'(x) = \frac{1}{x}$ $\quad g'(x) = 14x$ ← DIFFERENTIATE SEPARATELY

$f'(g(x)) = \frac{1}{7x^2}$

$\frac{dy}{dx} = f'(g(x))g'(x) = \left(\frac{1}{7x^2}\right)(14x) = \frac{2}{x}$ ← REASSEMBLE THE PIECES

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What are the special cases of the chain rule?

- It allows us to differentiate a function raised to a power:
  - $\frac{d}{dx}((f(x)^n) = n(f(x))^{n-1}f'(x)$

e.g. DIFFERENTIATE $y = (x^2 - 5x + 7)^7$ ← $n = 7$

$f(x) = x^2 - 5x + 7$

$f'(x) = 2x - 5$

$\frac{dy}{dx} = n(f(x))^{n-1}f'(x) = 7(x^2 - 5x + 7)^6(2x - 5)$

$\frac{dy}{dx} = 7(2x - 5)(x^2 - 5x + 7)^6$

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- It allows us to differentiate functions that are not given in the form **y= f(x):**
  - $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$

e.g. FIND THE VALUE OF $\frac{dy}{dx}$ AT THE POINT (3,1) ON THE CURVE WITH EQUATION $y^3 + 2y^2 = x$

$\frac{dx}{dy} = 3y^2 + 4y$ ← DIFFERENTIATE x WITH RESPECT TO y

$\frac{dy}{dx} = \frac{1}{\left(\frac{dx}{dy}\right)} = \frac{1}{3y^2 + 4y}$ ← INVERT TO FIND $\frac{dy}{dx}$

AT (3,1) $y = 1$ AND

$\frac{dy}{dx} = \frac{1}{3(1)^2 + 4(1)} = \frac{1}{7}$

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- There is a useful case which is used to integrate certain functions:
  - $\frac{d}{dx}(\ln(f(x)) = \frac{f'(x)}{f(x)}$

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straight (ie which function is ***y*** and which is ***u***, or which is ***f*** and which is ***g***).

Worked Example

For the curve with equation $y = 5^{x^4}$, find the gradient of the curve at the point $(1, 5)$.

Give your answer as an exact value.

DEFINE y AND u

$y = 5^u \qquad u = x^4$

DIFFERENTIATE SEPARATELY

$\frac{dy}{du} = (\ln 5)5^u \qquad \frac{du}{dx} = 4x^3$

REASSEMBLE

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = (\ln 5)5^u(4x^3)$

$\frac{dy}{dx} = (4\ \ln 5)x^3(5^{x^4})$

WHEN $x = 1$,

$\frac{dy}{dx} = (4\ \ln 5)(1)^3(5^{1^4}) = 20\ln 5$

THIS COULD BE DONE USING THE FUNCTION NOTATION VERSION OF THE CHAIN RULE, WITH $f(x) = 5^x$ AND $g(x) = x^4$

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